System Requirements

- Operating System: macOS
- CPU: ARM64 (Apple Silicon)
- Compiler: Apple clang 17.0.0 (clang-1700.4.4.1)
- Libraries: GNU Scientific Library (GSL)

Tested Environment

- macOS 15.6.1 (Sequoia)
- GSL 2.7.1

Run model calibration:

```
./exec -s 1 -rerun 1 -e .
```

Table 1: Command line arguments

Argument	Type	Description
-s	int	Random seed
-verbose	int	Parameter for diagnostic checks • 1: print diagnostic checks • 0: suppress them
-rerun	int	Controls whether to perform model calibration or run simulations using known parameters. Specifically: • `0`: perform model calibration using the Metropolis–Hastings MCMC procedure. • `!=0`: load parameter sets from a previous calibration and run model simulations.
-e	string	Path to the output directory

Table 2: Description of input and output files

File name	Dimension (rows x columns)[a]	Type	Sep	Description
INPUT				
parameters	3x2	**Column 1:** string **Column 2:** double/int	Tab	**Column 1:** parameter name **Column 2:** parameter value

		Parameter details: OMEGA: double GAMMA: double NSIM: int		Parameter details: • OMEGA: rate of progression from exposed to infectious individuals (weeks^{-1}) • GAMMA: recovery rate (weeks^{-1}) • NSIM: Number of simulations (same as number of rows of input file *mcmcrerun*)
mcmcrerun	NSIMx3	double	Tab	Parameter sets sampled from the posterior distribution obtained through MCMC. **Column 1:** scaling factor for transmissibility **Column 2:** reporting ratio **Column 3:** number of infectious individuals at the start of the simulation
ILI_A_2023_2024	Wx3	double	Tab	**Column 1:** week of the 2023/2024 influenza season **Column 2:** weekly incidence of influenza-like illnesses per 1,000 related to influenza A viruses
matrix_IT_presence	AxA	double	Tab	Average number of contacts of an individual attending work/school in-person of age group a (row) with individuals of age group $\tilde{a}$ (column)
matrix_IT_not_presence	AxA	double	Tab	Average number of contacts of an individual not attending work/school in-person of age group a (row) with individuals of age group $\tilde{a}$ (column)
pop_by_age	Ax1	Int	Tab	Population size by age group a (row) in the Italian population
pop_by_age_presence	Ax1	Int	Tab	Number of individuals of age a (row) attending work/school in-person
pop_by_age_not_presence	Ax1	Int	Tab	Number of individuals of age a (row) not attending work/school in-person
susc_age_input	Ax1	double	Tab	Relative susceptibility to infection of age group a compared to the reference group (a=1)
OUTPUT				
CumInfByAgeP_* * runs from 0 to NSIM-1	AxW	double	Tab	Number of new infections in individuals attending school/work in-person (age group a, week w)
CumInfByAgeNP_* * runs from 0 to NSIM-1	AxW	double	Tab	Number of new infections in individuals not attending school/work in-person (age group a, week w)

[a]: W: number of simulation weeks (24; from week 46 of 2023 to week 17 of 2024); A: number of age groups (15).